Testing and Validation Report

SEO Fixer · 30 October 2026 · Sohan Reddy

Test run result

All 31 tests passed in 2.3 seconds on 30 October 2026, and Django's system check found no issues.

- Command: `python manage.py test seofixer`
- No API keys are needed. Claude, GitHub and PageSpeed answers are simulated.
- HTML parsing, Chroma search, image processing and page requests run for real.

What the tests check

Group	Tests	What is proven
`ParsingTests`	7	Title, meta tags and headings are read correctly; empty alt on decorative images counts as valid; rules follow Google guidance on word count and JSON-LD; headings after the first 40 are still checked; `noindex` is matched as a word, not part of a word; text chunks keep their overlap and end; only public addresses are accepted; repo names are parsed correctly
`AgentTests`	8	An agent searches before it reports; no key means rule-only mode; an empty index gives a clear message; only verified sources and page text can be applied; web results become sources and paused searches resume; the index takes over when web search is off; all 4 agents really run at the same time; a report always comes back
`ImageAndEditTests`	7	A reused image uses its largest shown width; a text edit needs exactly one match across all files; image attributes are not added twice; image paths are matched only when certain; resizing keeps the format and saves bytes; the screenshot diff includes new page area; better/worse follows each metric's direction
`RetrievalTests`	1	Real Chroma cosine search works and a refresh removes old chunks
`GitHubTests`	3	One commit and one draft PR are built; image and loading edits land in the same commit; edits run only on a complete repo listing
`PageFlowTests`	5	The page loads and CSRF is enforced; a keyless audit keeps measurements and rules; a PR uses only saved server data and is opened once; only current forms and valid selections are accepted; measuring again keeps the first baseline

Validation built into the code

Every outside input is checked before it is used.

Input	Check	Where
Website URL	`http`/`https` only, no username or password, port 80 or 443, and every DNS address must be public	`audit.validate_url`
Every browser request	The same URL check on each request and redirect; websockets closed; service workers blocked	`audit.fetch_page`
Page response	Only successful pages (status under 400) are audited	`audit.fetch_page`
GitHub repo	Exactly `owner/repo` or a GitHub link	`github.parse_repo`
Form posts	GET/POST only; CSRF token; `audit_id` must match the saved audit; only the 4 known actions	`views.home`
Fix selection	Numbers must be in range; at least one change must be picked	`views.open_selected_pr`
AI report	Summary is text; score is a whole number 0 to 100; at most 4 fixes; severity is high, medium or low; fields under 4,000 characters	`ai.validate_report`
AI sources and edits	A source must come from the agent's own searches; edit text must exist on the page	`ai.validate_report`
AI tool calls	Bad calls go back to the agent to correct; queries cut to 500 characters; at most 3 searches	`ai.run_agent`
Repo edits	Text must match exactly once; complete repo listing; up to 150 files of up to 200 KB; same-site images under 25 MB	`github.py`
Image width	Must be a positive whole number	`github.optimize_image`

Screenshot ID	Must be 32 hex characters and the file must exist	`measurements.visual_difference`
Page text sent to AI	Treated as data, never as instructions	`prompts.FIX_RULES`

Test approach

- **Fast and free.** Claude, GitHub and Google answers are simulated, so all 31 tests run in about 2 seconds with no keys and no cost.
- **Real where it matters.** HTML parsing, Chroma search, image processing and page requests run for real.
- **Full flow covered.** Page-flow tests use sample measurements, so the audit, PR and re-measure steps are checked end to end without opening a browser.
- **One command.** `python manage.py test seofixer` runs everything.